

**Nº 1**

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```

ошибка: совпадений не найдено: quarto-cli
egorlisenkov@fedora:~/uni/lab08$ sudo dnf install -y git python3-notebook
[sudo] пароль для egorlisenkov:
Предупреждение: не удалось загрузить «/etc/yum.repos.d/rpmsfusion-free.repo», пропускается.
RPM Fusion for Fedora 39 - Nonfree - NVIDIA Driver                26 kB/s | 131 kB    00:05
Errors during downloading metadata for repository 'rpmsfusion-nonfree-nvidia-driver':
- Status code: 404 for https://mirrors.rpmfusion.org/metalink?repo=nonfree-fedora-nvidia-driver-39&arch=x86_64 (IP: 158.69.60.128)
- Status code: 404 for https://mirrors.rpmfusion.org/metalink?repo=nonfree-fedora-nvidia-driver-39&arch=x86_64 (IP: 78.47.223.143)
Error: Не удалось загрузить метаданные для репозитория «rpmsfusion-nonfree-nvidia-driver»: Cannot
repare internal mirrorlist: Status code: 404 for https://mirrors.rpmfusion.org/metalink?repo=nonfree-fedora-nvidia-driver-39&arch=x86_64 (IP: 78.47.223.143)
Oracle Linux / RHEL / CentOS - x86_64                          2.2 kB/s | 862 B    00:00
Oracle Linux / RHEL / CentOS - x86_64                          1.1 MB/s | 1.2 kB    00:00
Oracle Linux / RHEL / CentOS - x86_64                          2.2 kB/s | 862 B    00:00
Error: Не удалось загрузить метаданные для репозитория «virtualbox»: repomd.xml GPG signature verification error: Signing key not found
Игнорируется репозиторий: rpmsfusion-nonfree-nvidia-driver, virtualbox
Последняя проверка окончания срока действия метаданных: 0:00:41 назад, 8т 26 мая 2026 19:04:41.
Пакет git-2.47.0-1.fc39.x86_64 уже установлен.
Зависимости разрешены.

```

| Пакет                        | Архитектура | Версия        | Репозиторий | Размер |
|------------------------------|-------------|---------------|-------------|--------|
| Установка:                   |             |               |             |        |
| python3-notebook             | noarch      | 7.0.7-1.fc39  | updates     | 3.1 M  |
| Установка зависимостей:      |             |               |             |        |
| jupyterlab                   | noarch      | 4.0.11-1.fc39 | updates     | 5.8 M  |
| python-jupyter-filebrowser   | noarch      | 5.1.0-5.fc39  | fedora      | 9.7 M  |
| python3-anyio                | noarch      | 3.7.0-2.fc39  | fedora      | 215 K  |
| python3-appdirs              | noarch      | 1.4.4-11.fc39 | fedora      | 23 K   |
| python3-argon2-cffi          | noarch      | 21.3.0-4.fc39 | fedora      | 38 K   |
| python3-argon2-cffi-bindings | x86_64      | 21.2.0-4.fc39 | fedora      | 27 K   |
| python3-arrow                | noarch      | 1.2.3-4.fc39  | fedora      | 153 K  |
| python3-asttokens            | noarch      | 2.4.1-1.fc39  | updates     | 62 K   |
| python3-async-lru            | noarch      | 2.0.4-1.fc39  | fedora      | 21 K   |
| python3-babel                | noarch      | 2.12.1-6.fc39 | fedora      | 6.9 M  |
| python3-backcall             | noarch      | 0.1.0-20.fc39 | fedora      | 32 K   |
| python3-bleach               | noarch      | 6.1.0-1.fc39  | updates     | 64 K   |
| python3-brotli               | x86_64      | 1.1.0-1.fc39  | fedora      | 337 K  |
| python3-comm                 | noarch      | 0.1.4-1.fc39  | fedora      | 18 K   |
| python3-contourpy            | x86_64      | 1.2.0-1.fc39  | updates     | 265 K  |

[illegible]

# Julia

```
[4a2e7e0d] + Sockets
[fa267f1f] + TOML v1.0.3
[a4e569a6] + Tar v1.10.0
[cf7118a7] + UUIDs
[4ec0a83e] + Unicode
[deac9b47] + LibCURL_jll v8.4.0+0
[e37daf67] + LibGit2_jll v1.6.4+0
[29816b5a] + LibSSH2_jll v1.11.0+1
[c8ff49c3] + MbedTLS_jll v2.28.2+1
[14a3606d] + MozillaCACerts_jll v2023.1.10
[83775a58] + Zlib_jll v1.2.13+1
[8e850ede] + nghttp2_jll v1.52.0+1
[3f19e933] + p7zip_jll v17.4.0+2

Info Packages marked with x have new versions available but compatibility constraints restrict them from upgrading. To see why use `status --outdated -m`
Precompiling project...
 18 dependencies successfully precompiled in 48 seconds. 1 already precompiled.

(@v1.10) pkg> AC

julia> using DrWatson

julia> initialize_project("project"; authors = "Egor Lisenkov", force = true)
Activating new project at `~/uni/lab08/project`
Resolving package versions...
Updating `~/uni/lab08/project/Project.toml`
[634d3b9d] + DrWatson v2.19.1
Updating `~/uni/lab08/project/Manifest.toml`
[0b6fb165] + ChunkCodecCore v1.0.1
[4c0bbec4] + ChunkCodecLibZlib v1.0.0
[55437552] + ChunkCodecLibZstd v1.0.0
[634d3b9d] + DrWatson v2.19.1
[5789e2e9] + FileIO v1.18.0
[076d061b] + HashArrayMappedTries v0.2.0
[033835bb] + JLD2 v0.6.3
[692b3bcd] + JLLWrappers v1.7.1
[1914dd2f] + MacroTools v0.5.16
[bac558e1] + OrderedCollections v1.8.1
x [aea7be01] + PrecompileTools v1.2.1
[21216c6a] + Preferences v1.5.1
[ae029012] + Requires v1.3.1
[7e506255] + ScopedValues v1.5.0
[6c6a2e73] + Scratch v1.3.0
```

```
[c8ffd9c3] + MbedTLS_jll v2.28.2+1
[14a3606d] + MozillaCACerts_jll v2023.1.10
[83775a58] + Zlib_jll v1.2.13+1
[8e850ede] + nghttp2_jll v1.52.0+1
[3f19e933] + p7zip_jll v17.4.0+2

Info Packages marked with ⚠ have new versions available but compatibility constraints
restrict them from upgrading. To see why use `status --outdated -m`
"project"

julia> DrWatson.projectname()
"project"

julia> ^C
```

. 4: image/4.png

```

[0ac62f75] + libass_jll v0.17.4+0
[1183f4f0] + libdecor_jll v0.2.2+0
[2db6ffa8] + libevdev_jll v1.13.4+0
[f638f0a6] + libfdk_aac_jll v2.0.4+0
[36db933b] + libinput_jll v1.28.1+0
[b53b4c65] + libpng_jll v1.6.55+0
[f27f6e37] + libvorbis_jll v1.3.8+0
[009596ad] + mtdev_jll v1.1.7+0
[1270edf5] + x264_jll v10164.0.1+0
[dfaa095f] + x265_jll v4.1.0+0
[d8fb68d0] + xkbcommon_jll v1.13.0+0
[8ba89e20] + Distributed
[9fa8497b] + Future
[37e2e46d] + LinearAlgebra
[9abbd945] + Profile
[1a1011a3] + SharedArrays
[2f01184e] + SparseArrays v1.10.0
[10745b16] + Statistics v1.10.0
[4607b0f0] + SuiteSparse
[8dfed614] + Test
[e66e0078] + CompilerSupportLibraries_jll v1.1.1+0
[4536629a] + OpenBLAS_jll v0.3.23+4
[05823500] + OpenLibm_jll v0.8.1+2
[efcefd7] + PCRE2_jll v10.42.0+1
[bea87d4a] + SuiteSparse_jll v7.2.1+1
[8e850b90] + libblastrampoline_jll v5.8.0+1

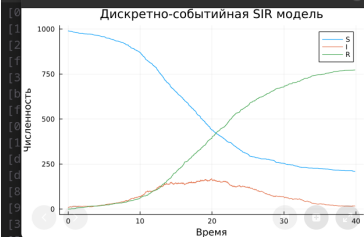
```

**Info** Packages marked with x have new versions available but compatibility constraints  
dict them from upgrading. To see why use `status --outdated -m`  
recompiling project...

213 dependencies successfully precompiled in 255 seconds. 27 already precompiled.

julia>

# SIR



```
[9ab8b945] + Profile
[1a1011a3] + SharedArrays
[2f01184e] + SparseArrays v1.10.0
[10745b16] + Statistics v1.10.0
[4607b0f0] + SuiteSparse
[8dfed614] + Test
[e66e0078] + CompilerSupportLibraries_jll v1.1.1+0
[4536629a] + OpenBLAS_jll v0.3.23+4
[05823500] + OpenLibm_jll v0.8.1+2
[efcefd7] + PCRE2_jll v10.42.0+1
[bea87d4a] + SuiteSparse_jll v7.2.1+1
[8e850b90] + libblastrampoline_jll v5.8.0+1

Info Packages marked with ⚠ have new versions available but compatibility constraints
dict them from upgrading. To see why use `status --outdated -m`
recompiling project...
213 dependencies successfully precompiled in 255 seconds. 27 already precompiled.

julia> exit()
gorlisenkov@fedora:~/uni/lab08$ julia scripts/sir_des.jl
gorlisenkov@fedora:~/uni/lab08$
```

```
gorlisenkov@fedora:~/uni/lab08$ cd ~/uni/lab08
julia

      _       _       _
     (_)_    (_)_    (_)_
    _ _      | _ _    | _ _
   | | | | | | | | | | | |
   | | | _ | | | | | | | |
   _/ | \_ ' _ | | | \_ ' _
  _/

Documentation: https://docs.julialang.org
Type "?" for help, "]?" for Pkg help.
Version 1.10.3 (2024-04-30)
Official https://julialang.org/ release

julia> using Pkg
julia> Pkg.add("IJulia")
ERROR: `using` is not a recognized command. Type ? for help with available commands

julia> using Pkg

julia> Pkg.add("IJulia")
Updating registry at `~/.julia/registries/General.toml`
Downloading [=> ] 0.1 % %
```

```
gorlisenkov@fedora:~/uni/lab08$ cd ~/uni/lab08
ulia
```

```

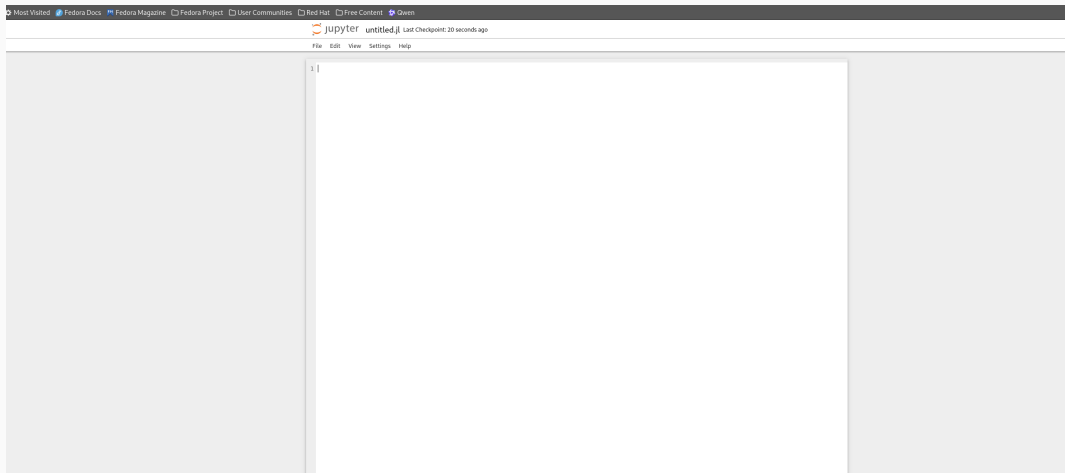
┌───┐ ┌───┐ ┌───┐ | Documentation: https://docs.julialang.org
└───┘ └───┘ └───┘ |
┌───┐ ┌───┐ ┌───┐ |
└───┘ └───┘ └───┘ |
┌───┐ ┌───┐ ┌───┐ | Type "?" for help, "]?" for Pkg help.
└───┘ └───┘ └───┘ |

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| 54 minutes ago |        |
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| 57 minutes ago | 146 B  |
| 57 minutes ago | 1005 B |
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```
[1]: include("src/sir_model.jl")

using Random, StatsPlots

[2]: tmax = 40.0
u0 = [990, 10, 0] # S, I, R
p = [0.05, 10.0, 0.25] # β, c, γ

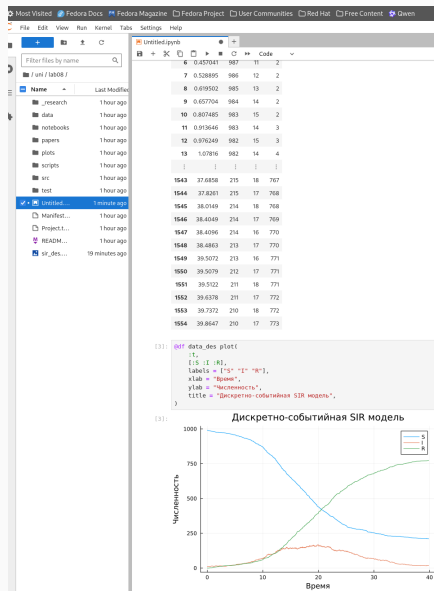
Random.seed!(1234)

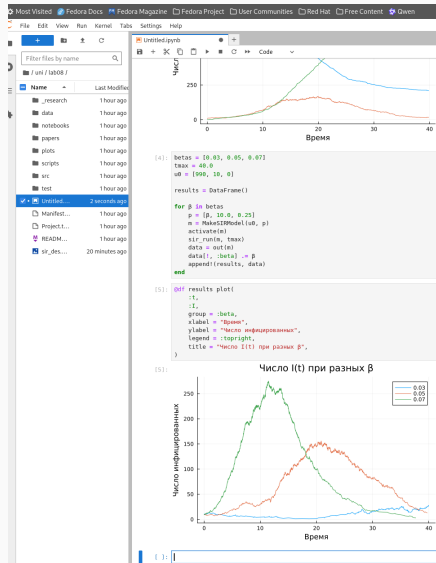
des_model = MakeSIRModel(u0, p)
activate(des_model)
sir_run(des_model, tmax)
data_des = out(des_model)

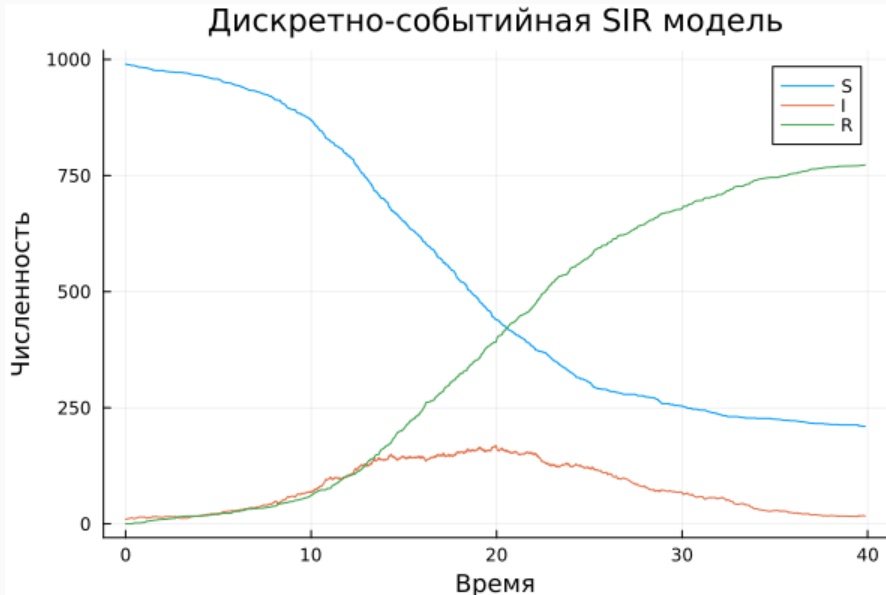
[3]: 1554x4 DataFrame
      Row  t      S      I      R
      ----  -  -  -  -
      Float64 Int64 Int64 Int64
1      0.0  990    10    0
2  0.110851  989    11    0
3  0.249585  988    12    0
4  0.374392  987    13    0
5  0.487738  987    12    1
6  0.457041  987    11    2
7  0.528895  986    12    2
8  0.619502  985    13    2
9  0.657704  984    14    2
10 0.807485  983    15    2
11 0.913646  983    14    3
12 0.976249  982    15    3
13 1.07816   982    14    4
⋮      ⋮      ⋮      ⋮      ⋮
1543 37.6858  215    18   767
1544 37.8261  215    17   768
1545 38.0149  214    18   768
1546 38.4049  214    17   769
1547 38.4096  214    16   770
1548 38.4863  213    17   770
1549 39.5072  213    16   771
1550 39.5079  212    17   771
1551 39.5122  211    18   771
1552 39.6378  211    17   772
1553 39.7372  210    18   772
1554 39.8647  210    17   773
```

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# SIR-









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